FRACTION STUDIO PROPORTIONAL TOWERS

4-Way Multi-Notational Fraction Blocks (10 Denominators: 1/1 to 1/12)

SERIES 1 • CARD 04 / 12 • DOMAIN: PROPORTIONAL TOWERS & MULTI-NOTATION EQUIVALENCE

CARD 04 / 12 • FRAC-01

MATHEMATICAL GUIDE

1. CONCEPT & GEOMETRIC PROOF

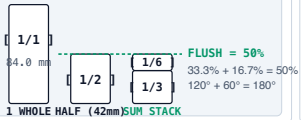
PROPORTIONAL HEIGHT & 4-WAY MULTI-NOTATION

Fractions are proportional lengths on a 7mm grid.
Each block face shows fraction, decimal, and percent:
[1/3] (Symbol) = 0.333 (Decimal) = 33.3% (Percent)
Equal heights prove mathematical equivalence!

UNIVERSAL 84.00 MM WHOLE DATUM

- 1 Whole Datum: [1/1] = 12U = 84.00 mm reference rod
- Unit Base: [1/12] = 1U = 7.00 mm (Matches [2])
- Sub-Units: [1/2]=42mm, [1/3]=28mm, [1/4]=21mm

PROPORTIONAL EQUIVALENCE: 1/3 + 1/6 = 1/2



2. KIT PARTS INVENTORY (65 PIECES)

COMPONENT	QTY	SPEC	COLOR
Whole [1/1]	2x	84.00 mm	PLA Tan
Halves [1/2]	4x	42.00 mm	PLA Tan
Thirds [1/3]	3x	28.00 mm	Muted Red
Fourths [1/4]	4x	21.00 mm	Matte Blue
Fifths [1/5]	5x	16.80 mm	PLA White
Sixths [1/6]	6x	14.00 mm	Muted Red
Eighths [1/8]	8x	10.50 mm	Matte Blue
Ninths [1/9]	9x	9.33 mm	Muted Red
Tenths [1/10]	10x	8.40 mm	PLA White
Twelfths [1/12]	12x	7.00 mm	Matte Blue
13U Box & Lid	2x	98.4 mm	PLA Gray/White

TOTAL: 63 Blocks (12 Columns × 84mm) + 13U Box + Stage Lid = 65 Parts

SLANT 3D PRODUCTION PALETTE

- PLA White: Base-10 fractions (5x 1/5, 10x 1/10) & Stage Lid.
- PLA Matte Blue: Binary/duodecimal family (4x 1/4, 8x 1/8, 12x 1/12).
- PLA Matte Muted Red: Ternary fraction family (3x 1/3, 6x 1/6, 9x 1/9).
- PLA Tan: Unit wholes & halves (2x 1, 4x 1/2) • PLA Gray: 13U Box Body.

3. PACK-UP & SUPPORT

✓ 10-SECOND PACK-UP

Stand all 63 blocks across the 12 floor sockets (2 cols of 1, 2 cols of 1/2, and 1 col each of 1/3..1/12). Every column levels at 84.0 mm; missing pieces leave an instant

✓ NO LOST PIECE GUARANTEE

Classrooms lose pieces. Every single math component is permanently stocked for single replacements.

stackcalc.io/parts
Order replacements anytime

✓ HANDS-ON TUTORIAL

Flip over for 5-step concrete lesson & live 3D Video QR!
Keep stored inside box lid!

FUNDAMENTAL THEOREM

Proportional height is strictly preserved under common denominator fraction addition.

STACKCALC ARCHIVAL INSTRUMENT LABS
Designed in Mountain View, California • Non-Toxic BPA-Free • Classroom Lab Stations: stackcalc.io/schools

OFFICIAL REFILL PORTAL
stackcalc.io/parts

LESSON GUIDE: COMMON DENOMINATOR ADDITION & EQUIVALENCE

Step-by-Step Hands-On Tutorial: 1/3 + 1/4 = 7/12 & 4-Way Multi-Notation

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CARD 04 / 12 • FRAC-01

LESSON TUTORIAL

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN
stackcalc.io/products/fraction-studio/
Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS 5.NF.A.1 (Unlike Denom...)
- CCSS 4.NF.A.2 (Fraction Com...)
- TEKS 111.7.b.3.H (Rational Ope...)

Grades 3–8 Fractions & Proportions

GUIDING INQUIRY

Which smaller block family levels off flush with both fraction towers in the comparison slot?

1 STEP 1: SEAT 1/1 WHOLE IN REFERENCE WELL

Place the 84.00 mm Whole [1/1] block in the left reference slot.
Its 4 faces show: 1/1, 1.00, 100%, and 360°.
This physical block establishes the 100% benchmark height.

2 STEP 2: STACK 1/3 AND 1/4 BLOCKS

Place one [1/3] block (28.00 mm) in the right comparison well.
Stack one [1/4] block (21.00 mm) directly on top of the [1/3].
Total combined height: 28.00 + 21.00 = 49.00 mm.

3-4 STEPS 3 & 4: TEST TWELFTHS & SOLVE

3. TEST COMMON TWELFTHS: Stack 7 twelfths: 7 × 7.00 mm = exactly 49.00 mm! Flush summit!
4. READ 4-WAY FACES: 0.333 + 0.250 = 0.583; 120° + 90° = 210°.
Calculator Verification: Press Fraction Display Icon ([FDISP] & [g][SIMP]) and run 1 [ENTER] 3 [÷] 1 [ENTER] 4 [÷] [+] [FDISP] → 7/12.

5 CORE PRINCIPLE: COMMON DENOMINATORS ARE COMMON SIZES

You cannot directly add thirds and fourths because their blocks are different physical heights!
Converting to twelfths (4/12 + 3/12 = 7/12) literally swaps disparate pieces for identical 7.00 mm unit blocks so their heights can be counted directly!

★ EXTENSION INVESTIGATION: ADD: 1/2 + 1/3

Seat [1/2] (42mm) and [1/3] (28mm) together. Test sixths: how many [1/6] blocks reach the exact 70.0mm summit? (Scan QR for 3D animated solution!)

→ Assemble with physical blocks, then scan QR code for 3D animated solution.

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PARTS & REFILL PORTAL
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STACKCALC STEM PEDAGOGY & PHYSICAL COMPUTING

CONCRETE • REPRESENTATIONAL • ABSTRACT

1. TACTILE CONCRETE

7.00mm 1U PITCH
Every physical block dimension is locked to an exact integer ratio, providing tactile feedback of quantity.

MOTOR KINESTHETIC
Sliding, stacking, and interlocking builds spatial memory and intuitive number sense before symbols appear.

CONSERVATION OF QUANTITY
Mass, area, and volume remain invariant regardless of how blocks are grouped or rotated.

SELF-CORRECTING
Precision mechanical detents seat flush only when arithmetic identities and geometric proofs are correct.

CCSS.MATH.PRACTICE.MP5
Use appropriate tools strategically.

2. HARDWARE RPN STACK

POSTFIX ARITHMETIC
[ARG 1] [ENTER] [ARG 2] [OP]. Zero operator ambiguity, zero parentheses, and zero hidden precedence rules.

X, Y, Z, T REGISTERS
Every intermediate result is visible on the display stack. What is stacked on the desk is preserved in memory.

DIGITAL TWIN OF TOWERS
Physically stacking blocks onto a stage lid directly mirrors pushing values into the digital stack registers.

STEP-BY-STEP REASONING
Students observe each operator execute in real time, making compound calculations completely transparent.

CCSS.MATH.PRACTICE.MP1
Make sense of problems & persevere.

3. LEARNING CONTINUUM

GRADES K–2: FOUNDATION
Ten-Frames, Number Rack & Dominoes for subitizing, counting, 5-and-10 anchors, and place value fluency.

GRADES 3–5: FRACTIONS
Fraction bars, prime factorization towers, time gears, and discrete combinatorics with probability dice.

GRADES 6–12: ADVANCED
Algebraic factoring rails, unit circle volvelles, complex numbers, and professional scientific RPN workflows.

DIFFERENTIATED TIERS
Tactile physical manipulates bridge English language learners, visual learners, and accelerated STEM students.

CCSS.MATH.PRACTICE.MP4
Model with mathematics.

CLASSROOM CARE & SANITIZATION

1. FOOD-SAFE RECYCLED POLYMERS:
Manufactured from high-purity virgin polymer (PLA / PETG). Non-toxic, odorless, BPA-free, and compliant with RoHS, EN71-3, and ASTM F963 safety standards.

2. CLASSROOM CLEANING & DISINFECTION:
Wash blocks in warm water with mild liquid dish soap (up to 55°C / 130°F) or wipe with standard classroom disinfectant wipes. Allow to air dry flat before storage.

3. MODULAR STORAGE DETENTS:
Stackable 3x1 and 3x3 stage lids feature multi-axis detents tested to 10,000+ opening cycles. Trays pack rattle-free for classroom mobile cart transport.

4. SOCRATIC CLASSROOM DISCUSSION:
Encourage students to physically construct models to test hypotheses before executing calculator calculations: "What invariant property did your physical model prove?"

FRAC-01: IN-BOX SPECIFICATION & QC AUDIT

CALIFORNIA LAB QC

CALIFORNIA LAB QC AUDIT CHECKLIST

[✓] MITUTOYO DIGITAL CALIPER DATUM:
All block dimensions conform to ±0.05 mm tolerance across universal 7.00 mm 1U visible height system.

[✓] 1.85 MM RETENTION SOCKET INTERFERENCE:
Stacking plugs seat flush with 0.15 mm diametral interference for zero-wobble tower verticality.

[✓] ZERO-RATTLE DETENT LOCK:
Multi-axis lid detents lock securely over 3x1 bin perimeter rim. Audible snap with 1.4 N retention.

[✓] CONTINUOUS 360° MICRO-FLUTING & SEAM TRAP:
2.0 mm pitch, 0.50 mm flutes. Slicer seam locked into rear-left corner seam trap notch.

[✓] FOOD-SAFE RECYCLED POLYMER:
100% virgin PLA / PETG. RoHS and EN71-3 compliant, non-toxic, odorless, child-safe.

[✓] IN-BOX COLLATERAL PLACEMENT:
Place this reference card inside the box lid or inside the storage tray.

RPN HARDWARE COMPATIBILITY

- Compatible with StackCalc SC-6 (Early STEM)
- Compatible with StackCalc SC-32 (Scientific RPN)
- Zero parentheses, zero operator precedence errors
- Physical towers are the exact digital twin of the stack!

TOTAL KIT INVENTORY: 65 PIECES

Verified by Assembly Tech & Machine Vision at Packing Bench.

Inspection Pass • Mountain View, CA

LIFETIME REPLACEMENT GUARANTEE

NO LOST PIECE LEFT BEHIND™

Classrooms lose blocks. We understand. Every individual manipulative block has a permanent part number. Visit stackcalc.io/parts to order exact replacement pieces with same-day dispatch ($1.99–$2.99/piece).

PORTAL: stackcalc.io/parts

DISTRICT & CLASSROOM PROCUREMENT

- Net-30 Invoicing for US Public & Private School Districts.
- Volume Classroom Bundles (10-packs & 30-packs).
- Free Teacher Activity Books & Lesson Blueprints.
- Title I STEM & Carl D. Perkins Grant eligible.

Email: procurement@stackcalc.io
stackcalc.io/schools

3D INTERACTIVE VIDEO DEMOS

Scan the QR code on the back of this card (Side B) to access:

- 60fps 3D animated walkthrough of each lesson
- Interactive web 3D model with explode view
- Step-by-step calculator keystroke simulations
- Printable student worksheets and answer keys

VIDEOS: stackcalc.io/products

Designed & Engineered in Mountain View, California